

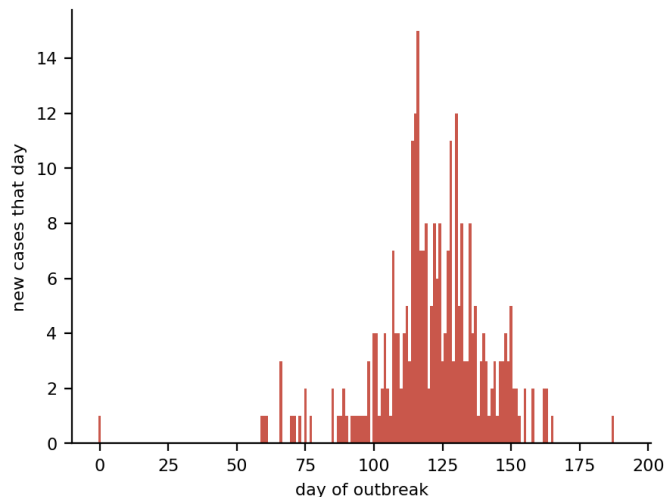
Ebola — Kikwit, DR Congo, 1995

292 cases over six months in a city hospital and its surrounding community

People at risk	Days recorded	Days usable	Total cases	Deaths	Died of those infected
~1,000	192	139	292	236	81%

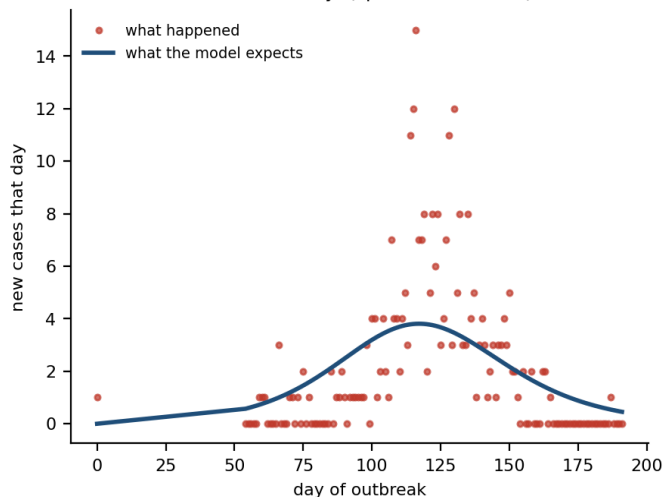
This shows that this was a severe but contained outbreak: four in five people who caught it died, which is why it ended after a few hundred cases rather than spreading further.

The outbreak over time



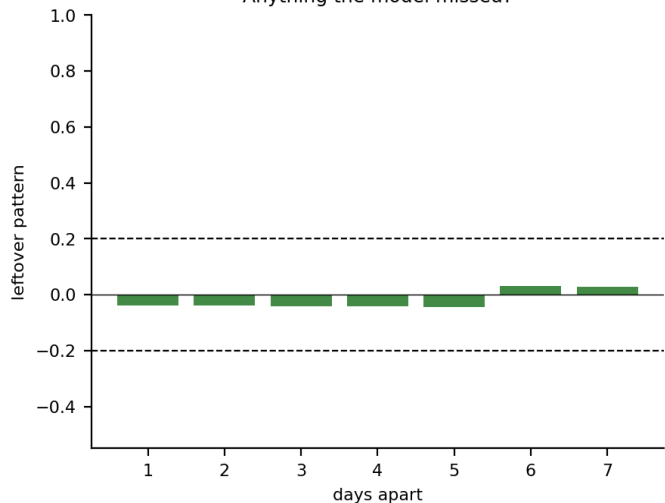
This shows that cases climbed for about three months and then fell away. One rise and one fall means a single wave that ran out of people to infect — not a disease that kept coming back.

Model vs reality (spread score 1.92)



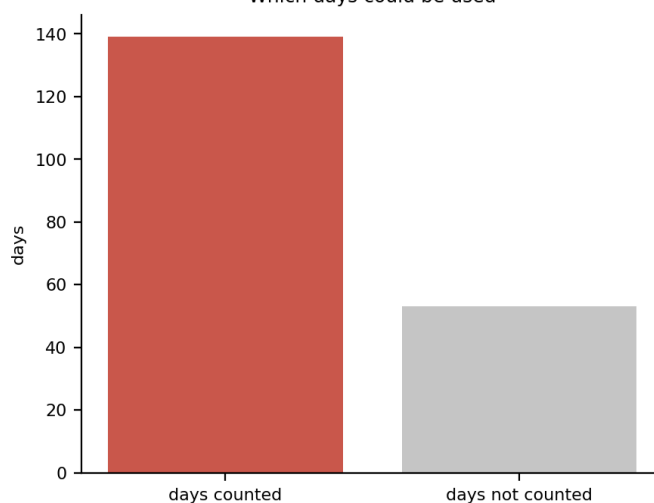
This shows that the line is what our model expects; the dots are what was actually recorded. They follow the same arc, so the assumptions we made about how long people take to fall ill and how long they can pass it on are close to right.

Anything the model missed?



This shows that after the model explains what it can, almost nothing is left over — the bars are tiny and green. If the model had missed something that changed over time, these bars would be tall and red.

Which days could be used



This shows that 53 of the 192 days had no active case-finding, so we set those aside instead of treating them as days with genuinely zero cases. Counting them as real zeros would make the outbreak look smaller than it was.

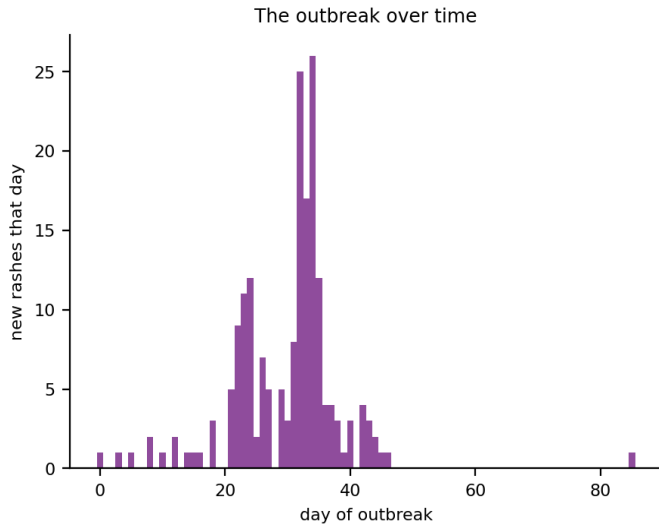
VERDICT: usable as-is. The model matches the shape of the outbreak, leaves no unexplained pattern behind, and the contagiousness it estimates sits inside the range other published studies of this same outbreak report.

Measles — Hagelloch, Germany, 1861

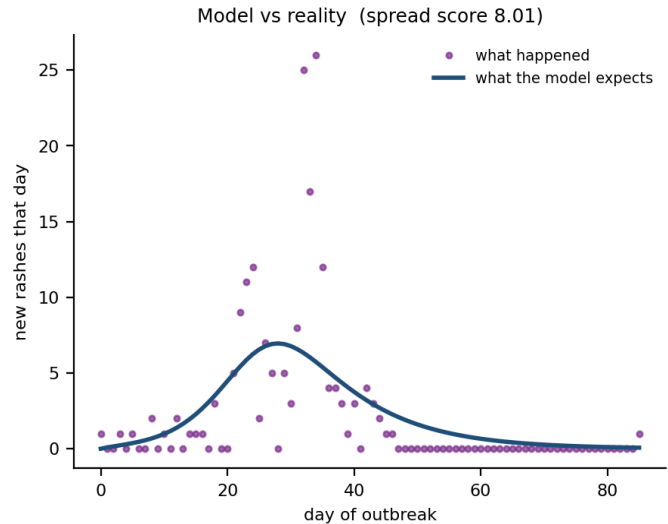
Every child in one village, recorded door to door by the local physician

Children in village	Days recorded	Children infected	Deaths	Died of those infected	School classes
188	86	188 (all of them)	12	6.4%	3

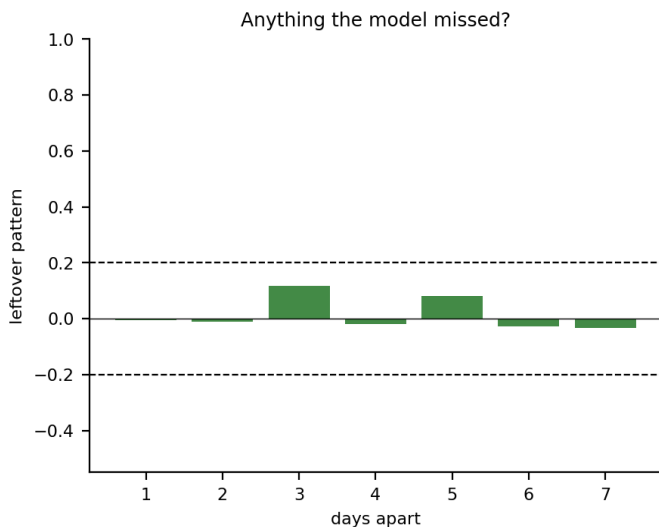
This shows that every single child in the village caught measles. That is what an extremely contagious disease looks like when it reaches a group with no prior immunity and no vaccine.



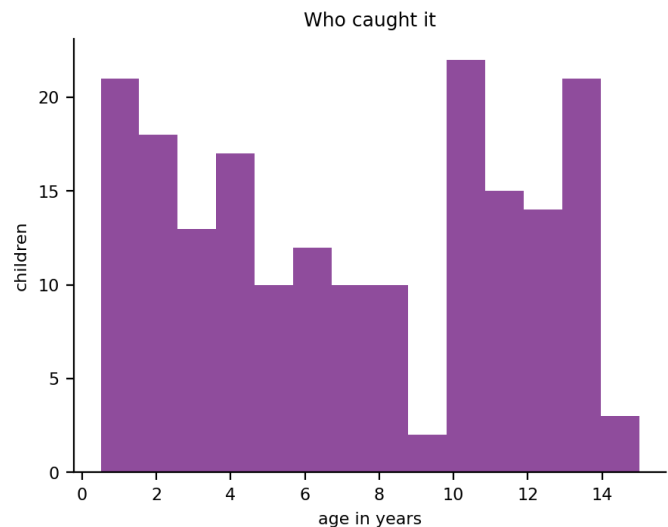
This shows that a slow trickle for a month, then a sharp burst, then it was over. The burst is the schoolrooms: once it reached a classroom it went through the whole class in days.



This shows that the model follows the timing well but draws a smoother, lower hump than reality. That gap is the classroom effect — our model assumes everyone mixes evenly, and in a village they do not.



This shows that nothing systematic is left over — the bars are small and green. The model may smooth out the burst, but it is not getting the timing or the direction wrong.



This shows that this hit children of every age from infants to fifteen-year-olds. Nobody was spared, which confirms there was no existing immunity in the village to slow it down.

VERDICT: usable as-is. Clean timing, nothing unexplained left behind, and the one imperfection — a smoother curve than reality — is a known and harmless consequence of assuming everyone mixes evenly in a village with schools.

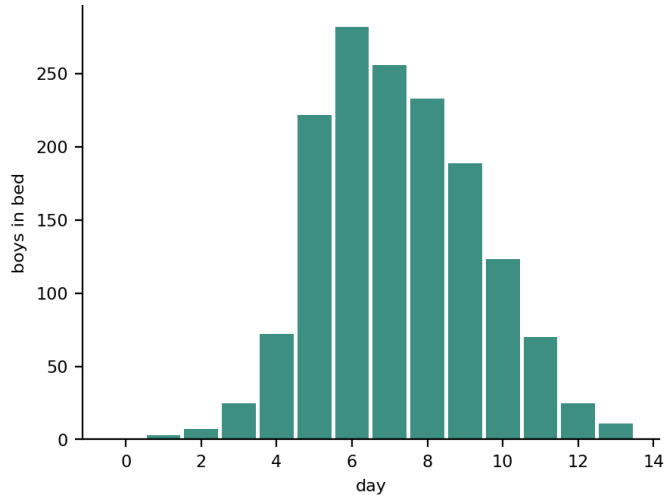
Influenza — English boarding school, 1978

A closed school of 763 boys; the matron counted who was in bed each morning

Boys at risk	Days recorded	Most in bed at once	Total who fell ill	Share who fell ill	Deaths
763	14	282 (37%)	512	67%	0

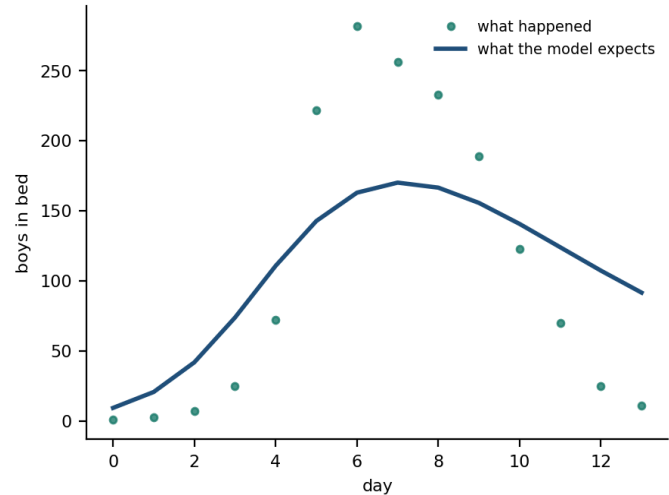
This shows that two thirds of the school caught flu inside a fortnight and nobody died — a very contagious but mild illness, the opposite profile to Ebola.

The outbreak over time



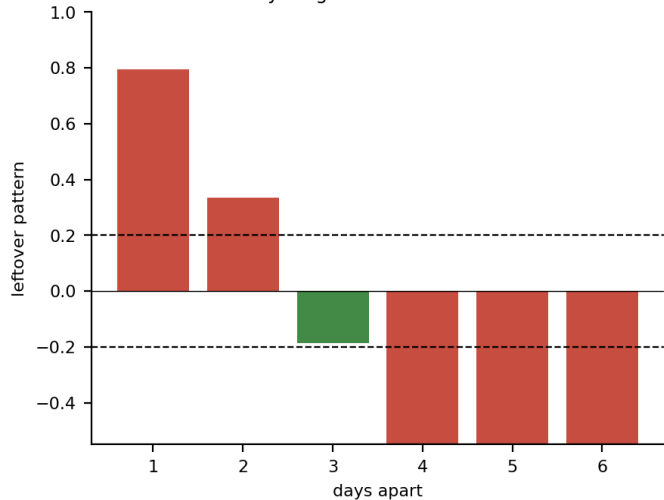
This shows that the whole thing was over in two weeks: up from one boy to 282 in six days, then back down. This is the fastest of the four outbreaks by a wide margin.

Model vs reality (spread score 6.35)



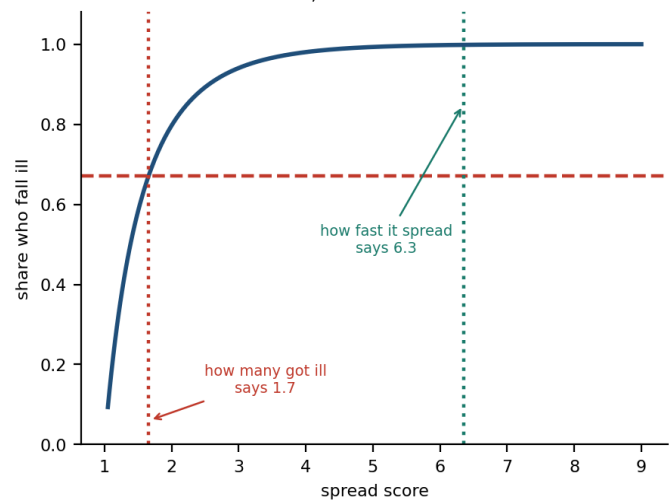
This shows that the model gets the rough shape but is visibly too flat — it cannot rise as fast or fall as fast as the real thing did. This is the first sign of trouble.

Anything the model missed?



This shows that a large red bar means a real pattern is still left over after the model has done its best. Something about this outbreak our model cannot capture.

Two clues, two different answers



This shows that the two ways of judging contagiousness disagree. Counting how many eventually fell ill points to a low number; watching how fast it spread points to a much higher one. Both cannot be right.

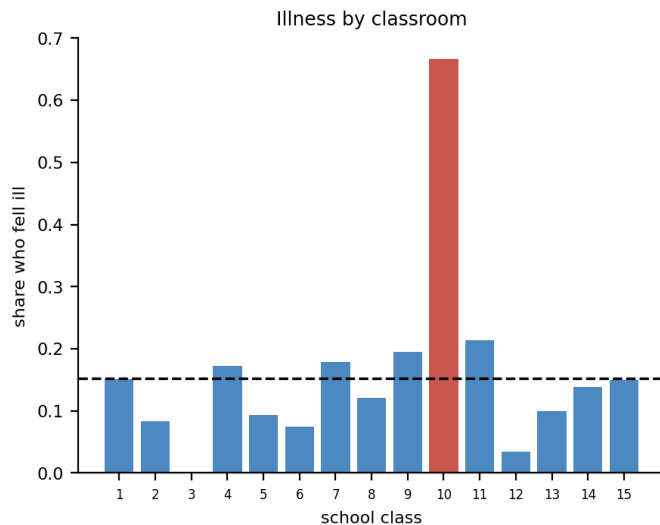
VERDICT: keep, but label it. This is a well-known dataset that standard models cannot fully explain — published attempts reach the same impasse. It is useful as a test of whether our checks catch a problem, and they did. It should not be quoted as a clean contagiousness estimate alongside the other three.

Norovirus — Derbyshire primary school, England, 2001

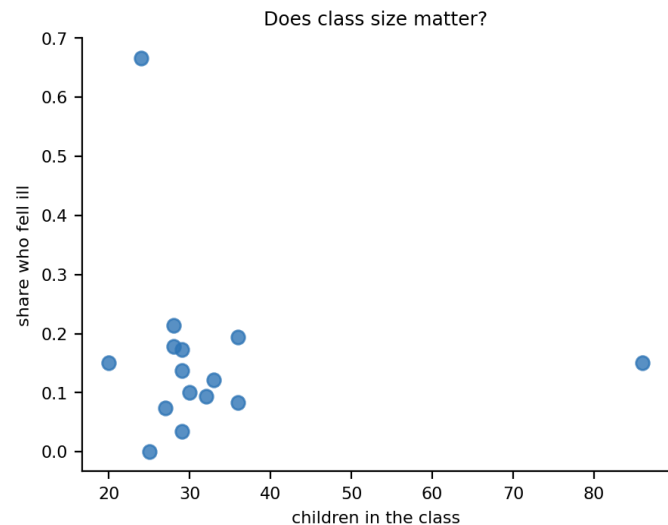
A stomach-bug outbreak recorded child by child across 15 classrooms

Children	Classes	Fell ill	Share who fell ill	Worst class	Class with none ill
492	15	75	15%	Class 10 (67%)	Class 3 (0%)

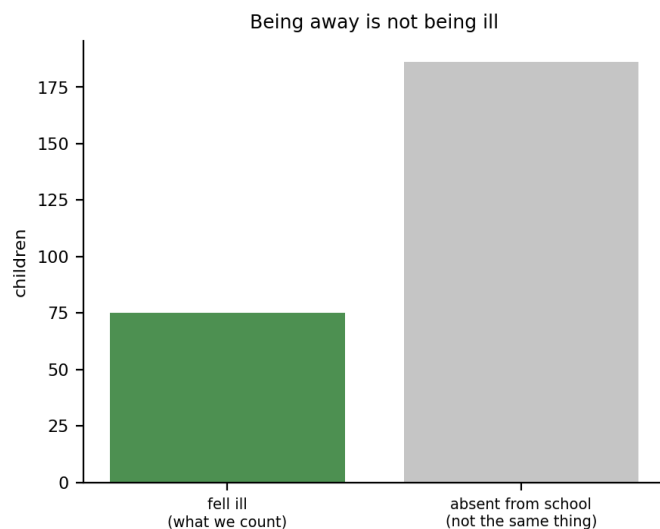
This shows that only one child in seven fell ill, and it was very uneven between classrooms — which is the clue that where a child sat mattered as much as the bug itself.



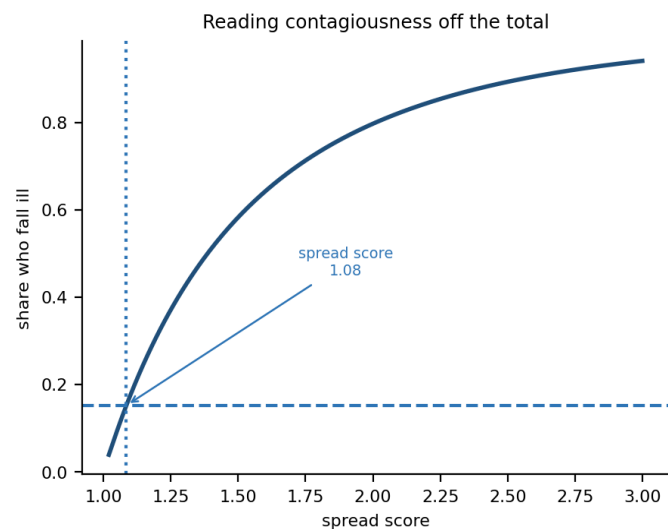
This shows that one classroom had two thirds of its children ill while another had none. An outbreak that spreads by close contact shows up as clusters like this, not as an even sprinkling across the school.



This shows that big classes were not worse hit than small ones, so the clustering is about who sat near whom, not simply about crowding.



This shows that far more children were absent than were actually ill. If we had counted absence as illness we would have more than doubled the outbreak. We count only children recorded as unwell.



This shows that for a one-off outbreak there is a fixed relationship between how contagious something is and what share of a group eventually catches it. We read our estimate off that curve using the 15% who fell ill.

VERDICT: usable as-is, and the most precise of the four. Every child was recorded individually, so we have nearly 500 separate observations rather than a daily tally — the classroom differences are handled explicitly rather than ignored.